

Smart Electricity Consumption Prediction

Energy Optimization System — Project Report

1. Dataset Summary

Original synthetic dataset created for this project (no public dataset used).

Total Records: 1200 | Total Features: 22

Target Variable: Daily_Electricity_Consumption_kWh

Mean Consumption: 17.35 kWh/day

Std Deviation: 9.44 kWh

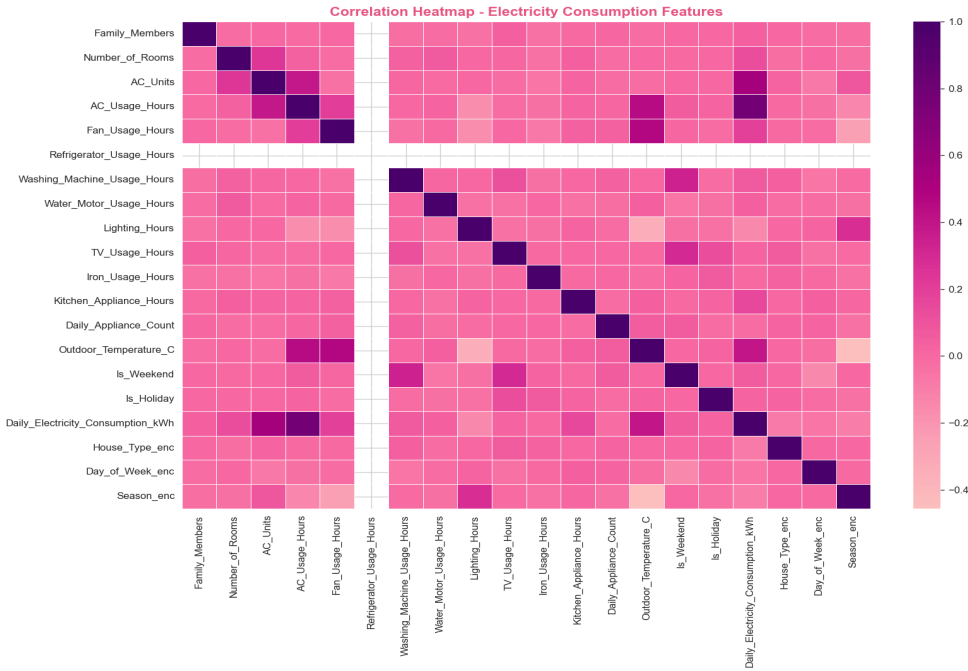
2. Model Performance Report

Best Performing Model: **Gradient Boosting**

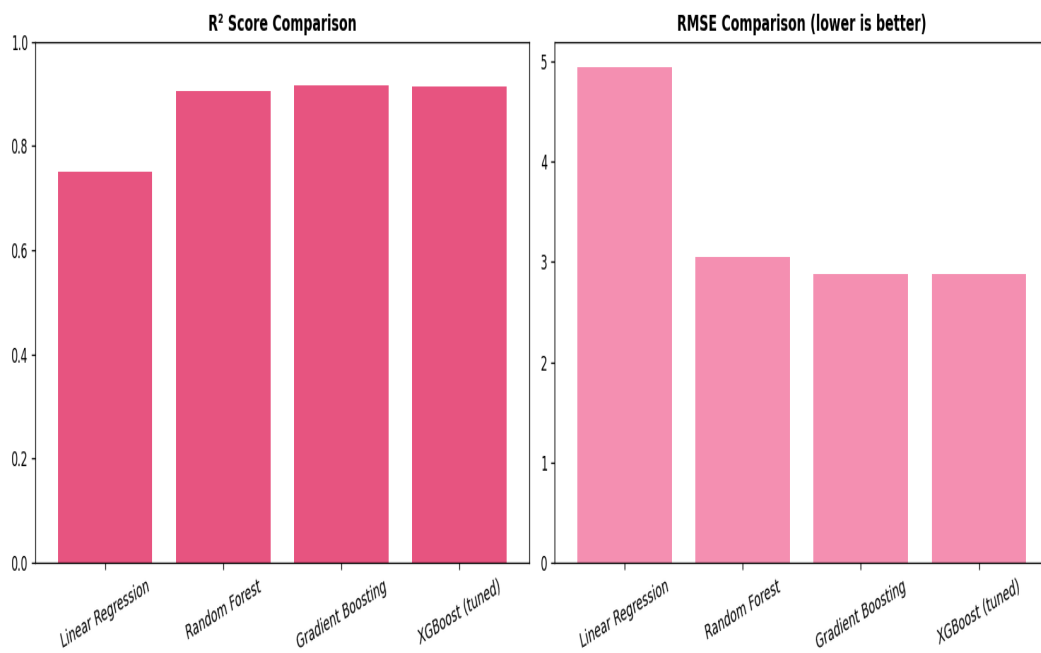
Model	MAE	MSE	RMSE	R2 Score
Linear Regression	3.916	24.472	4.947	0.7512
Random Forest	2.055	9.287	3.047	0.9056
Gradient Boosting	1.905	8.295	2.88	0.9157
XGBoost (tuned)	1.907	8.315	2.884	0.9154

3. Exploratory Data Analysis

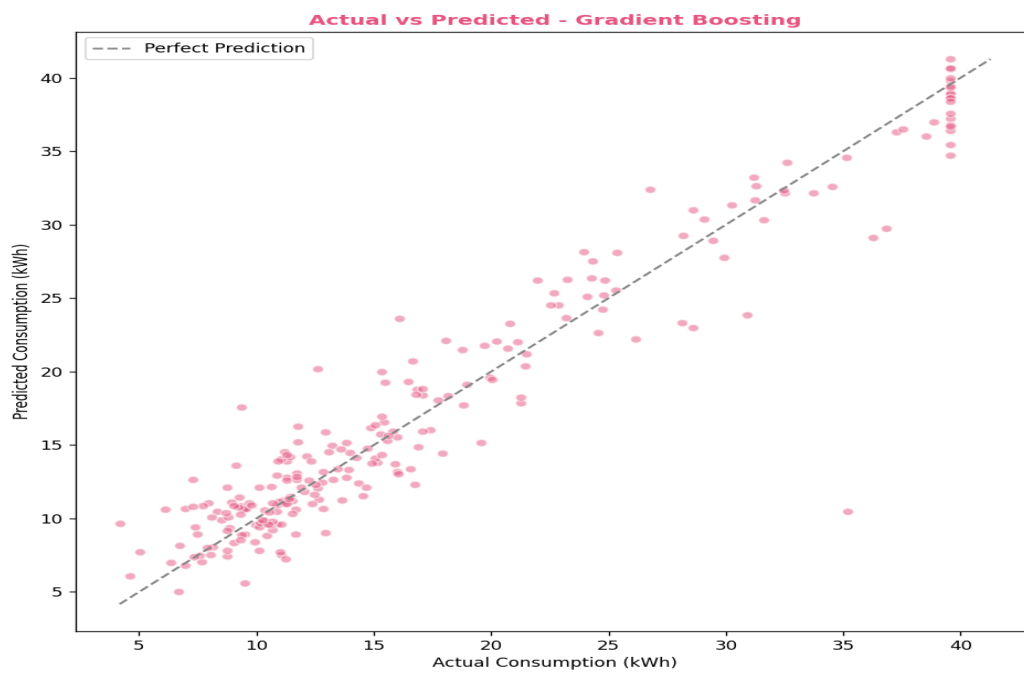
Correlation Heatmap



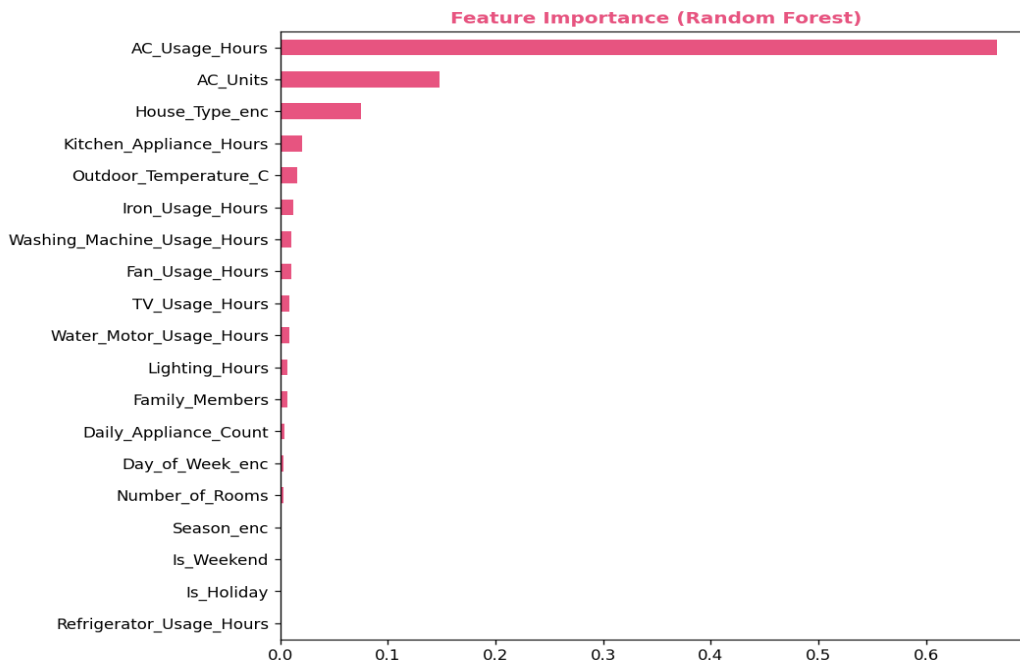
Model Comparison (R2 / RMSE)



Actual vs Predicted Consumption



Feature Importance



4. Energy Consumption Insights

EXPLORATORY DATA ANALYSIS - BUSINESS INSIGHTS

Dataset size after cleaning: 1200 records, 23 columns

Average daily consumption: 17.35 kWh

Median daily consumption: 13.96 kWh

Std deviation: 9.44 kWh

Top factors INCREASING electricity consumption:

- AC_Usage_Hours: correlation = 0.786
- AC_Units: correlation = 0.544
- Outdoor_Temperature_C: correlation = 0.393

Factors most associated with LOWER consumption:

- Season_enc: correlation = -0.096
- Lighting_Hours: correlation = -0.143
- Refrigerator_Usage_Hours: correlation = nan

Highest consuming season: Summer (avg 23.00 kWh/day)

Lowest consuming season: Spring (avg 12.41 kWh/day)

Highest consuming house type: Bungalow (avg 24.11 kWh/day)

Business Implication: AC usage and outdoor temperature are the dominant drivers of consumption, suggesting energy-saving campaigns should focus on cooling appliance efficiency during summer/monsoon months.

5. Energy Optimization Recommendations

- Shift heavy appliance usage (AC, washing machine, iron) to off-peak hours (before 5 PM or after 10 PM).
- Households using AC more than 6 hrs/day can save 15-25% by raising thermostat by 2°C.
- Bungalows and independent houses show the highest consumption — insulation upgrades recommended.
- Summer/Monsoon months require proactive energy budgeting due to cooling load spikes.

- Standby/base load (chargers, routers) contributes a steady baseline — smart plugs recommended.

6. Challenges Faced & Future Improvements

Challenges: Designing a realistic synthetic dataset that reflects genuine appliance-usage correlations (e.g., AC hours vs temperature) without access to real smart-meter data required careful domain-driven simulation rather than random generation.

Future Improvements: Integrate a live Weather API for real-time temperature-based prediction, deploy via Docker for production use, add real smart-meter data collection, and extend the model to multi-day/weekly forecasting.